

Humanoid Robot ‘ Digit ’ – Whole-Body Control Foundation Model | NVIDIA

Source HTML: 机器人产品资料库_2026-06-21/02_Word补充资料/企业与产品资料/062_NVIDIA_case_study_-_Agility_Robotics_Digit.html

Visit your regional NVIDIA website for local content, pricing, and where to buy partners specific to your country.

- Products Cloud Services Creating Data Center Embedded Systems Gaming Graphics Cards and GPUs Laptops and Desktops Networking Professional Workstations Software Tools Cloud Services DGX Cloud NVIDIA ’ s AI factory in the cloud NVIDIA APIs Explore, test, and deploy AI models and agents Private Registry Guide for using NVIDIA NGC private registry with GPU cloud NVIDIA NGC Accelerated, containerized AI models and SDKs DSX Platform Build AI factories optimized for lowest cost tokens per megawatt Creating NVIDIA Studio High performance GeForce RTX PCs, purpose-built for creators NVIDIA Broadcast App AI-enhanced voice and video for next-level streams, videos, and calls NVIDIA App and Studio Drivers Optimize gaming, streaming, and AI-powered creativity RTX AI PCs AI PCs for gaming, creating, productivity and development RTX Remix Create RTX remasters of classic games with open-source AI Project G-Assist AI assistant to optimize and control your GeForce RTX PC Data Center Overview Modernizing data centers with AI and accelerated computing DGX Platform Enterprise AI factory for model development and deployment Grace CPU Architecture for data centers that transform data into intelligence HGX Platform A supercomputer purpose-built for AI and HPC IGX Platform Advanced functional safety and security for edge AI MGX Platform Accelerated computing with modular servers OVX Systems Scalable data center infrastructure for high-performance AI DSX Platform Build AI factories optimized for lowest cost tokens per megawatt Embedded Systems Jetson Leading platform for autonomous machines and embedded applications DRIVE AGX Powerful in-vehicle computing for AI-driven autonomous vehicle systems IGX Platform Advanced functional safety and security for edge AI Gaming GeForce Explore graphics cards, gaming solutions, AI technology, and more GeForce Graphics Cards RTX graphics cards bring game-changing AI capabilities Gaming Laptops Thinnest and longest lasting RTX laptops, optimized by Max-Q G-SYNC Monitors Smooth, tear-free gaming with NVIDIA G-SYNC monitors DLSS Neural rendering tech boosts FPS and enhances image quality Reflex Ultimate responsiveness for faster reactions and better aim RTX Remix Create RTX remasters of classic games with open-source AI Project G-Assist AI assistant to optimize and control your GeForce RTX PC GeForce NOW Cloud Gaming RTX-powered cloud gaming. Choose from 3 memberships NVIDIA App and Game Ready Drivers Optimize gaming, streaming, and AI-powered creativity NVIDIA Broadcast App AI-enhanced voice and video for next-level streams, videos, and calls SHIELD TV World-class streaming media performance Graphics Cards and GPUs GeForce Graphics Cards RTX graphics cards bring game-changing AI capabilities NVIDIA RTX PRO Accelerating professional AI, graphics, rendering and compute workloads Virtual GPU Virtual solutions for scalable, high-performance computing Blackwell Architecture The engine of the new industrial revolution Hopper Architecture High performance, scalability, and security for every data center Ada Lovelace Architecture Performance and energy efficiency for endless possibilities Laptops and Desktops GeForce Desktops RTX graphics cards bring game-changing AI capabilities GeForce Laptops GPU-powered laptops for gamers and creators Studio Laptops High performance laptops purpose-built for creators NVIDIA RTX Spark PCs Slim laptops and small desktops with NVIDIA AI and RTX graphics RTX AI PCs AI PCs for gaming, creating, productivity and development NVIDIA RTX PRO Laptops Accelerate professional AI and visual computing from anywhere Networking Overview Accelerated networks for modern workloads DPUs Software-defined hardware accelerators for networking, storage, and security Ethernet Ethernet performance, availability, and ease of use across a wide range of applications InfiniBand High-performance networking for super computers, AI, and cloud data centers Networking Software Networking software for optimized performance and scalability Network Acceleration IO subsystem for modern, GPU-accelerated data centers Professional Workstations Overview Accelerating professional AI, graphics, rendering, and compute workloads DGX Spark A Grace Blackwell AI Supercomputer on your desk DGX Station The ultimate deskside AI supercomputer powered by NVIDIA Grace Blackwell DGX Station for Windows The world ’ s most powerful deskside AI supercomputer for Windows NVIDIA RTX PRO Desktops Powerful AI, graphics, rendering, and compute workloads NVIDIA RTX PRO Laptops Accelerate professional AI and visual computing from anywhere Software Agentic AI Models - Nemotron AI Agents - NeMo AI Blueprints AI Inference - Dynamo AI Inference - NIM AI Microservices - CUDA-X Automotive - DRIVE Data Science - Apache Spark Data Science - RAPIDS Decision Optimization - cuOpt Healthcare Platforms Industrial AI - Omniverse Intelligent Video Analytics - Metropolis NVIDIA AI Enterprise NVIDIA Mission Control NVIDIA Run:ai Physical AI - Cosmos Robotic ...

NVIDIA ’ s AI factory in the cloud

Explore, test, and deploy AI models and agents

Guide for using NVIDIA NGC private registry with GPU cloud

Accelerated, containerized AI models and SDKs

Build AI factories optimized for lowest cost tokens per megawatt

High performance GeForce RTX PCs, purpose-built for creators

AI-enhanced voice and video for next-level streams, videos, and calls

Optimize gaming, streaming, and AI-powered creativity

AI PCs for gaming, creating, productivity and development

Create RTX remasters of classic games with open-source AI

AI assistant to optimize and control your GeForce RTX PC

Modernizing data centers with AI and accelerated computing

Enterprise AI factory for model development and deployment

Architecture for data centers that transform data into intelligence

A supercomputer purpose-built for AI and HPC

Advanced functional safety and security for edge AI

Accelerated computing with modular servers

Scalable data center infrastructure for high-performance AI

Leading platform for autonomous machines and embedded applications

Powerful in-vehicle computing for AI-driven autonomous vehicle systems

Explore graphics cards, gaming solutions, AI technology, and more

RTX graphics cards bring game-changing AI capabilities

Thinnest and longest lasting RTX laptops, optimized by Max-Q

Smooth, tear-free gaming with NVIDIA G-SYNC monitors

Neural rendering tech boosts FPS and enhances image quality

Ultimate responsiveness for faster reactions and better aim

RTX-powered cloud gaming. Choose from 3 memberships

World-class streaming media performance

Accelerating professional AI, graphics, rendering and compute workloads

Virtual solutions for scalable, high-performance computing

The engine of the new industrial revolution

High performance, scalability, and security for every data center

Performance and energy efficiency for endless possibilities

GPU-powered laptops for gamers and creators

High performance laptops purpose-built for creators

Slim laptops and small desktops with NVIDIA AI and RTX graphics

Accelerate professional AI and visual computing from anywhere

Accelerated networks for modern workloads

Software-defined hardware accelerators for networking, storage, and security

Ethernet performance, availability, and ease of use across a wide range of applications

High-performance networking for super computers, AI, and cloud data centers

Networking software for optimized performance and scalability

IO subsystem for modern, GPU-accelerated data centers

Accelerating professional AI, graphics, rendering, and compute workloads

A Grace Blackwell AI Supercomputer on your desk

The ultimate desktide AI supercomputer powered by NVIDIA Grace Blackwell

The world ' s most powerful desktide AI supercomputer for Windows

Powerful AI, graphics, rendering, and compute workloads

Simplify AI development with NVIDIA AI Workbench on GPUs

Explore NVIDIA's AI models, blueprints, and tools for developers

Monitor and manage GPU performance in cluster environments

Explore NVIDIA developer tools for AI, graphics, and HPC

Discover GPU-optimized AI, HPC, and data science software

Optimize enterprise GPU management

Accelerate AI and HPC workloads with NVIDIA GPU Cloud solutions

Enhance multi-display productivity with NVIDIA RTX Desktop Manager

Creative tools and AI-powered apps for artists and designers

AI-powered audio and video enhancement

- Solutions Artificial Intelligence Cloud and Data Center Design and Simulation High-Performance Computing Robotics and Edge AI Autonomous Vehicles Artificial Intelligence Overview Add intelligence and efficiency to your business with AI and machine learning Agentic AI Build AI agents designed to reason, plan, and act AI Data Powering a new class of enterprise infrastructure for AI Conversational AI Enables natural, personalized interactions with real-time speech AI Cybersecurity AI-driven solutions to strengthen cybersecurity and AI infrastructure Data Science Iterate on large datasets, deploy models more frequently, and lower total cost Inference Drive breakthrough performance with AI-enabled applications and services Data Processing Data engines ready for AI Cloud and Data Center Overview Powering AI, HPC, and modern workloads with NVIDIA AI Storage Bringing codesigned enterprise storage into the era of agentic AI AI Factory Full-stack infrastructure for scalable AI workloads AI Grid Scale AI across connected, distributed AI infrastructure Accelerated Computing Accelerated computing uses specialized hardware to boost IT performance Cloud Computing On-demand IT resources and services, enabling scalability and intelligent insights Colocation Accelerate the scaling of AI across your organization Networking High speed ethernet interconnect solutions and services Sustainable Computing Save energy and lower cost with AI and accelerated computing Virtualization NVIDIA virtual GPU software delivers powerful GPU performance Design and Simulation Overview Streamline building, operating, and connecting metaverse apps Computer Aided-Engineering Develop real-time interactive design using AI-accelerated real-time digital twins Digital Twin Development Harness the power of large-scale, physically-based OpenUSD simulation Rendering Bring state-of-the-art rendering to professional workflows Robotic Simulation Innovative solutions to take on your robotics, edge, and vision AI challenges Scientific Visualization Enables researchers to visualize their large datasets at interactive speeds Vehicle Simulation AI-defined vehicles are transforming the future of mobility Extended Reality Transform workflows with immersive, scalable interactions in virtual environments High-Performance Computing Overview Discover NVIDIA ' s HPC solutions for AI, simulation, and accelerated computing HPC and AI Boost accuracy with GPU-accelerating HPC and AI Scientific Visualization Enables researchers to visualize large datasets at interactive speeds Simulation and Modeling Accelerate simulation workloads Quantum Computing Fast-tracking the advancement of scientific innovations with QPUs Robotics and Edge AI Overview Innovative solutions to take on robotics, edge, and vision AI challenges Robotics GPU-accelerated advances in AI perception, simulation, and software Edge AI Bring the power of NVIDIA AI to the edge for real-time decision-making solutions Vision AI Transform data into valuable insights using vision AI AI Grid Scale AI-native services efficiently across connected, distributed AI infrastructure Autonomous Vehicles Overview AI-enhanced vehicles are transforming the future of mobility Open Source AV Models and Tools For reasoning-based AV systems AV Simulation Explore high-fidelity sensor simulation for safe autonomous vehicle development Reference Architecture Enables vehicles to be L4-ready Infrastructure Essential data center tools for safe autonomous vehicle development In-Vehicle Computing Develop automated driving functions and immersive in-cabin experiences Safety State-of-the-art system for AV safety, from the cloud to the car

Add intelligence and efficiency to your business with AI and machine learning

Build AI agents designed to reason, plan, and act

Powering a new class of enterprise infrastructure for AI

Enables natural, personalized interactions with real-time speech AI

AI-driven solutions to strengthen cybersecurity and AI infrastructure

Iterate on large datasets, deploy models more frequently, and lower total cost

Drive breakthrough performance with AI-enabled applications and services

Data engines ready for AI

Powering AI, HPC, and modern workloads with NVIDIA

Bringing codesigned enterprise storage into the era of agentic AI

Full-stack infrastructure for scalable AI workloads

Scale AI across connected, distributed AI infrastructure

Accelerated computing uses specialized hardware to boost IT performance

On-demand IT resources and services, enabling scalability and intelligent insights

Accelerate the scaling of AI across your organization

High speed ethernet interconnect solutions and services

Save energy and lower cost with AI and accelerated computing

NVIDIA virtual GPU software delivers powerful GPU performance

Streamline building, operating, and connecting metaverse apps

Develop real-time interactive design using AI-accelerated real-time digital twins

Harness the power of large-scale, physically-based OpenUSD simulation

Bring state-of-the-art rendering to professional workflows

Innovative solutions to take on your robotics, edge, and vision AI challenges

Enables researchers to visualize their large datasets at interactive speeds

AI-defined vehicles are transforming the future of mobility

Transform workflows with immersive, scalable interactions in virtual environments

Discover NVIDIA 's HPC solutions for AI, simulation, and accelerated computing

Boost accuracy with GPU-accelerating HPC and AI

Enables researchers to visualize large datasets at interactive speeds

Accelerate simulation workloads

Fast-tracking the advancement of scientific innovations with QPUs

Innovative solutions to take on robotics, edge, and vision AI challenges

GPU-accelerated advances in AI perception, simulation, and software

Bring the power of NVIDIA AI to the edge for real-time decision-making solutions

Transform data into valuable insights using vision AI

Scale AI-native services efficiently across connected, distributed AI infrastructure

AI-enhanced vehicles are transforming the future of mobility

For reasoning-based AV systems

Explore high-fidelity sensor simulation for safe autonomous vehicle development

Enables vehicles to be L4-ready

Essential data center tools for safe autonomous vehicle development

Develop automated driving functions and immersive in-cabin experiences

State-of-the-art system for AV safety, from the cloud to the car

- Industries Industries Overview Architecture, Engineering, Construction & Operations Automotive Cybersecurity Energy Financial Services Healthcare and Life Sciences Higher Education Game Development Government Manufacturing Media and Entertainment Restaurants Retail and CPG Robotics Semiconductor Smart Cities Supercomputing Telecommunications
- Shop
- Drivers
- Support
- US
- Sign In NVIDIA Account NVIDIA Store Account Logout
- Log In LogOut
- Login LogOut
- Shop Drivers Support

Manufacturing

Humanoid Robot ‘ Digit ’ Powered by Whole-Body Control Foundation Model

Agility Robotics

Objective

Agility Robotics is a developer of general-purpose humanoid robots that can work in various environments. Digit, the company ’ s humanoid robot, is designed to work seamlessly in human-built environments without costly facility modification. Digit ’ s initial use cases focus on addressing critical labor shortages in manufacturing, warehouses, and distribution centers. Ultimately, Agility Robotics intends to deploy Digit across use cases from industrial settings to homes and hotels. To overcome the inherent complexity of achieving reliable whole-body control for humanoid robots operating safely alongside human workers, Agility Robotics leveraged NVIDIA Isaac Sim™ and Isaac™ Lab, reference robotics simulation and learning frameworks built on NVIDIA Omniverse™, to create a breakthrough training methodology.

Partner

Customer

GXO Logistics

Schaeffler

Use Case

Robotics

Products

NVIDIA Omniverse NVIDIA Isaac

Key Benefits

- Improved Step-Recovery Performance: Trained across billions of simulation steps to enhance stability management for Digit when encountering physical disturbances like sudden forces or unexpected terrain.
- Accelerated Development Timeline: Iteration cycles for developing and testing new controllers are reduced from weeks to days through multi-GPU physics-based simulation before real-world deployment.
- Enhanced Real-World Deployment Success: Successfully deployed trained whole-body control foundation models from simulation to real warehouse and manufacturing operations with improved reliability and task performance.

Engineering Complex Whole-Body Control, at Scale

U.S. warehouses face persistent labor gaps on material-handling lines, yet remodeling facilities for fixed automation is costly and slow. Humanoid robots offer a drop-in alternative, but only if they can move with human-level agility in spaces never designed for machines.

The Solution

Agility Robotics takes a “simulation-first” approach to teaching its Digit robot how to stay on its feet in new environments.

Engineers begin by modeling the robot’s chassis in Isaac Sim, using OpenUSD to preserve every joint, mass, and contact surface. Customer CAD and BIM data drop into the same scene graph, and the team scripts thousands of stress-test scenarios—varying aisle widths, floor friction, lighting, and shove forces—to probe every balance-breaking edge case.

Digital twin simulations flow into Isaac Lab, where millions of parallel reinforcement-learning episodes refine Digit’s whole-body controller—the robot’s software “motor cortex.” Perception data stays in the loop, letting Digit learn how vision and motion interact. NVIDIA GPUs push training into the billions of interactions that true dexterity demands.

To prove the controller’s physics isn’t simulator-specific, Agility runs the same policy in a containerized MuJoCo pipeline; differences in contact physics expose corner cases and harden the policy.

By the time the code is deployed to the robot’s hardware, Digit can shrug off bumps, avoid obstacles, and get to work—without costly trial-and-error learning on the warehouse floor.

Scaling Humanoid Robot Intelligence for Enterprise Applications

Digit robots are now successfully deployed, performing complex material-handling tasks in live operations.

GXO Logistics: World’s First Humanoid RaaS Deployment Under an industry-first Robots-as-a-Service agreement, GXO is running Digit fleets in a Georgia fulfillment center. The robots pick totes, sort, and manage inventory alongside human crews, validating that skills learned in Isaac Sim carry straight into high-volume warehouse work.

Schaeffler: Precision Tasks in Automotive Manufacturing The global motion-technology supplier Schaeffler uses the Mega NVIDIA Blueprint to test workflows before hardware ships. At Schaeffler’s Cheraw, South Carolina plant, Digit now loads and unloads washing-machine housings—proving that a single whole-body foundation model can handle delicate, high-tolerance jobs once thought beyond general-purpose robots.

These rollouts show that Digit’s OpenUSD-trained skills scale from logistics totes to metal stampings, giving Agility and NVIDIA a fast-growing partner ecosystem—and confirming that humanoid intelligence is ready for enterprise production.

Looking Forward: The Future of AI-Powered Humanoid Robotics

Agility Robotics envisions a near future in which humanoid robots like Digit become everyday teammates—stocking shelves in retail aisles, shuttling parts on manufacturing floors, and filling labor gaps in warehouses and last-mile hubs. To reach that goal, the company is developing adaptive learning systems that let each robot watch a task demonstrated once, then practice in simulation to master it autonomously.

By training larger, more capable whole-body control foundation models in GPU-powered Isaac Sim—and validating them in Omniverse digital twins—Agility expects Digit to adapt to fresh workflows or unfamiliar layouts in hours rather than weeks. As models generalize across industries and environments, deploying humanoid robots will shift from bespoke integration projects to a scalable, plug-and-play reality for widespread use in almost any environment.

“Isaac Sim running on NVIDIA GPUs lets us simulate years of real-world learning for Digit in just hours. That simulation speedup means we can train for all the conditions we might see on the factory floor.”

Pras Velagapudi Chief Technology Officer, Agility Robotics

Learn more about accelerating automation with NVIDIA robotics solutions.

Related Customer Stories

- About Us
- Investors
- Venture Capital (NVentures)

- [NVIDIA Foundation](#)
- [Research](#)
- [Corporate Sustainability](#)
- [Technologies](#)
- [Careers](#)
- [Newsroom](#)
- [Company Blog](#)
- [Technical Blog](#)
- [Webinars](#)
- [Stay Informed](#)
- [Events Calendar](#)
- [GTC AI Conference](#)
- [NVIDIA On-Demand](#)
- [Developers](#)
- [Partners](#)
- [Executive Insights](#)
- [Startups and VCs](#)
- [NVIDIA Connect for ISVs](#)
- [Documentation](#)
- [Technical Training](#)
- [Professional Services for Data Science](#)
- [Privacy Policy](#)
- [Your Privacy Choices](#)
- [Terms of Service](#)
- [Accessibility](#)
- [Corporate Policies](#)
- [Product Security](#)
- [Contact](#)